

Part is part (plus pragmatics)

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Prologue

- ¹**Thesis:** *part* and its translational equivalents can be represented with the mereological notion of PROPER PART ($\sqsubset$)
- This may seem trivial, but several high-profile theories for parthood and plurality in natural language take this to not be the case
- This is partly due to the kinds of interpretations we can get for nominals like (1)

(1) part of the New Yorkers
- I argue instead that the canonical uses of (1) is actually a pragmatic effect, and a simple semantics for *part* is both tenable and desirable

1 Mereology as a theory of pluralities

- [Link \(1983\)](#): (definite) plural DPs are individuals (type e) rather than sets (type $\langle e, t \rangle$, cf. [Bennett 1974](#); [van der Does 1991](#); [Winter 2001](#))
- In lieu of sets, pluralities are mereological sums of individuals, whereas singulars are treated as atoms (i.e., individuals with no proper parts)
- A number of analytical advantages...
 - Avoid positing ambiguity for predicates ($\langle e, t \rangle \sim \langle \langle e, t \rangle, t \rangle$)
 - Maintain parity between plurals and mass nouns w.r.t collective predicates ($\{the\ boys/the\ water\}$ *gathered*), monotonicity, etc.
- Borrowing heavily from Classical Extensional Mereology ([Lesniewski, 1931](#); [Leonard and Goodman, 1940](#))
- To understand what it means to be a sum, we must first understand parthood; even the simplest mereologies assume the following maxims ([Varzi, 2003](#))

- (2) *Reflexivity*
 $\forall x. x \sqsubset x$
 (Everything is a part of itself.)

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(3) **Transitivity**

$$\forall x, y, z. (x \sqsubseteq y \wedge y \sqsubseteq z) \rightarrow x \sqsubseteq z$$

(Part of part of a whole is itself part of that whole.)

(4) **Antisymmetry**

$$\forall x, y. (x \sqsubseteq y \wedge y \sqsubseteq x) \rightarrow x = y$$

(A pair of objects which are parts of each other cannot be distinct objects.)

- We can define the notion of proper part as parthood plus a constraint against identity

$$(5) \quad x \sqsubset y \stackrel{\text{def}}{=} x \sqsubseteq y \wedge x \neq y$$

- Two objects are a sum when, for any object the sum is a part of, those two are also parts of said object

$$(6) \quad \text{Sum}(z, \{x, y\}) \stackrel{\text{def}}{=} \forall w. z \sqsubseteq w \leftrightarrow (x \sqsubseteq w \wedge y \sqsubseteq w) \quad (\text{Eberle 1967, a.o.})$$

- Going forward, I'll use $\oplus$ for the sum of two individuals and $\bigoplus$ for sums of sets with cardinalities > 2
- Parthood can conversely be illustrated as in (7)

$$(7) \quad x \oplus y = y \leftrightarrow x \sqsubseteq y$$

- A (definite) plural DP is the generalized sum of the set of NP

$$(8) \quad [[\text{New Yorker}]]^c = \{j(\text{erry}), g(\text{eorge}), e(\text{laine}), k(\text{ramer})\}$$

$$(9) \quad [[\text{the New Yorkers}]]^c = \bigoplus [[\text{New Yorker}]]^c = j \oplus g \oplus e \oplus k$$

- Definite plural is decomposable to a *-operator and maximality function
- Link's * takes a set of individuals and returns that set plus every possible sum of individuals from the base set

$$(10) \quad *P \stackrel{\text{def}}{=} \{x : \exists P' \subseteq P. x = \bigoplus P'\}$$

- A set taken by * is partially-ordered by mereological relation, the top being the highest-ranked element; thus, max will select the maximal sum

$$(11) \quad \max(P) \stackrel{\text{def}}{=} \iota x \in P. \neg \exists y \in P. x < y, \text{ otherwise undefined}$$

- Being a member of a plurality is now represented as $\sqsubset$ rather than $\subset$ or $\in$
- Recall that singular individuals are treated as atoms; the notion of minimal parts of a plurality is useful for, e.g., distributivity
- Yet, singularities are obviously composed of something
- Link divides parthood into two distinct relations: one which holds between pluralities and the singularities/subpluralities which compose them (ATOMIC part, $\sqsubset_A$), and one which holds between singularities and their subatomic parts (MATERIAL part, $\sqsubset_M$)

$$(12) \quad \text{a. } \text{Jerry} \sqsubset_A, \not\sqsubset_M \text{ the New Yorkers}$$

$$\text{b. } \text{Jerry's hand} \not\sqsubset_A, \sqsubset_M \text{ Jerry}$$

2 Indeterminacy of *part* and the transitivity problem

- Link's model is at odds with a strong cross-linguistic generalization: many partitive constructions can take both singular and plural DPs, the result of the first denoting a part of a singular individual and the second denoting a subplurality of a plurality (13)-(14)²

(13) part of the {book/books}

(14) ein Teil {des Apfels / der Äpfel}
 a part {the.GEN.SG apple.GEN.SG / the.GEN.PL apple.GEN.PL}
 '(a) part of the apple/apples' [German]

- This fact, observed by Moltmann (1997); Wagiel (2021), is hard to square if Link is correct
- The Link approach is salvageable if we allow *part* to be ambiguous as in (15)...

(15) a. $[[part_A]] = \lambda y \lambda x. x \sqsubset_A y$
 b. $[[part_M]] = \lambda y \lambda x. x \sqsubset_M y$

- ...but *part* does not pass the zeugma test—a diagnostic for ambiguity (Zwicky and Sadock, 1975; Lasersohn, 1995)

The zeugma test works as follows: A putatively ambiguous item is placed in construction with a coordinate structure. If it is authentically ambiguous, then it must take the same reading with respect to both conjuncts, or else it takes on the flavor of a joke. But if it is merely non-specific, it may be understood differently with respect to the different conjuncts, with no joke effect at all.

Lasersohn 1995, 93

- ...presumably because an ellided phrase is required to match its antecedent in semantics, not just phonological material
- (16) illustrates the diagnostic with English *bank*, which is ambiguous between a financial institution and a portion of land adjacent to a river

(16) ?John cashed a check at one bank and caught a fish at another.

- Yet (17) and its English equivalent exhibits no such effect, suggesting that *part* is not polysemous

(17) ein Teil der Birnen und des Apfels sind verfault
 a part DET.GEN pears.GEN and DET.GEN apple.GEN are rotten
 'Part of the pears and the apple are rotten' (Wagiel, 2021, §2.2.1)

- Given the above, Moltmann and Wagiel collapse the two relations into one; now, what holds between Jerry's hand + Jerry, Jerry + the New Yorkers is simply $\sqsubset$
- **We now have a problem, though:** recall the axiom (3) which established parthood as being a transitive relation
- If this is also true of natural language *part*, we would expect material parts of a singularity (e.g., Jerry's hand) to be part of pluralities which said singularity is also a member
- This prediction is not substantiated; for example, a scenario where Jerry's right hand and nothing else is covered in paint is not a verifying case for (18)

²See also partitives with other quantifiers (*some, all, most, half*, etc.).

(18) Part of the New Yorkers are covered in paint.

- Link's account, which I call SPLIT PARTHOOD (SP), sidesteps this issue entirely; different kinds of parthood for different kinds of objects, so transitivity has no bearing
- Maintaining a single parthood relation for parts of singularities and parts of pluralities requires some further assumptions
- Moltmann and Wagiel's strategy is to deviate from CEM's formulation of parthood and let transitivity fail under certain circumstances
- Specifically, they posit that for any given pair individuals, one can only be part of another if there is no intervening individual which is an INTEGRATED WHOLE (19)

(19) $\forall x, z. x \sqsubset z \rightarrow \neg \exists y. x \sqsubset y \wedge y \sqsubset z \wedge \text{int-wh}(y)$

- What is an IW? The two authors take different positions
- Moltmann (1997, §1.5) appeals specifically R-integrated wholes (Simons, 1987)

(20) An entity x is an R-integrated whole if there is a division of x such that every member of that division stands in the relation R to every other member and no member bears R to anything other than members of the division.

- Wagiel's definition for an IW is an individual who satisfies the mereotopological property of being MAXIMALLY-STRONGLY SELF-CONNECTED (MSSC, Casati and Varzi 1999)³
- Essentially, an MSSC individual is one who is SELF-CONNECTED (any two ways of dividing an individual are connected), has an interior which is self-connected, and the self-connected things which overlap with it are part of it
- A little more formally...

(21) Self-connected

$$\text{SC}(x) \stackrel{\text{def}}{=} \forall y \forall z [\forall w (\text{O}(w, x) \leftrightarrow (\text{O}(w, y) \vee \text{O}(w, z))) \rightarrow \text{C}(y, z)]$$
(x is self-connected iff any two parts that form the whole of x are connected to each other.)

(22) Strongly self-connected

$$\text{SSC}(x) \stackrel{\text{def}}{=} \text{SC}(x) \wedge \text{SC}(\text{INT}(x))$$
(x is strongly self-connected if x is self-connected and the interior of x is self-connected.)

(23) Maximally-strongly self-connected

$$\text{MSSC}(x) \stackrel{\text{def}}{=} \text{SSC}(x) \wedge \forall y [\text{SSC}(y) \wedge \text{O}(y, x) \rightarrow y \sqsubseteq x]$$

- Moltmann and Wagiel take different paths to derive the same intuition: a materially-real, single, whole individual like *Jerry* constitutes an integrated whole and (18) cannot be true when only Jerry's hand is covered in paint
- I call this analysis of parthood FORMALIZED TRANSITIVITY SUSPENSION (FTS)

3 Proposal

- FTS let's us have our cake and eat it, too: we need not stipulate ambiguity for *part*, and we can avoid the problematic prediction with subatomic parts and pluralities

³Specifically, an individual who is MSSC relative to a property.

- However, the condition on transitivity further separates natural language *part* and the metaphysical relation $\sqsubset$
- SP shares this trait, of course, but in a different way
- The problem I outline in §3 seems to present good motivation for this partition
- The present concern is whether it is appropriate in this case to enrich the semantics, or if the interpretation in (18) is resolvable to some other module
- **I propose that it is;** specifically, I argue that the conventional meaning of a partitive DP.PL is pragmatic (i.e., the result of competition)
- Consider (24) (without transitivity suspension)

$$(24) \quad \text{Parthood-Subset Maxim} \\ \forall a, b. (a \sqsubset b) \rightarrow (\{x : x \sqsubset a\} \subset \{y : y \sqsubset b\})$$

- Assuming a gunky universe (one where every individual has a proper part), both sets will have a cardinality of ∞
- Yet, *a*'s parts will constitute a proper subset of *b*'s parts because *b* contains *a* (and thus *a*'s parts are included in the set of *b*'s parts)
- Its relevance to the issue is particularly clear if we imagine *a* as a singular individual like *Cooper* and *b* as a plurality of which *a* is a member (*the linguists*)
- On its literal interpretation *part of Cooper* is far more informative than *part of the linguists*
- Compare a canonical case of scalar implicature with *some* $\sim$ *all*

$$(25) \quad \{w : \text{George ate } \forall \text{ eclaires in } w\} \subset \{w' : \text{George ate } \exists \text{ eclaires in } w'\}$$

- By conversational reasoning à la Grice (1975), a cooperative speaker will choose the least vague thing amongst competitors if they have the appropriate knowledge (MAXIM OF QUANTITY)
- Ergo, if a speaker wants to talk about a subatomic part of a singularity, they should use partitive DP.SG (even though partitive DP.PL literally contains the meaning of partitive DP.SG)
- In turn, if a listener hears partitive DP.PL, they can reason that the speaker did not mean partitive DP.SG, strengthening the meaning of the plural to an exclusive interpretation

4 Auxiliary support: inclusive interpretations

- This proposal makes a prediction about how plural partitives can be interpreted: items which have a conventional meaning derived by strengthening have been shown to exhibit the original (“weaker”) meaning in certain environments
- Plurals are a famous case (Sauerland 2003; Sauerland et al. 2005; et seq.); In questions or under universal quantifiers (i.e., cases of epistemic uncertainty), the inclusive reading of the plural emerges

(26) Q: Do you have cats?
A: Yes, I have one.

(27) Every man who has cats will attend the gala
 $\searrow$ Even men with only one cat will attend the gala

- For some speakers, partitive DP.PL exhibits the exact same behavior in these environments

(28) Q: Has Jerry painted part of the boats?

A: Yes, in fact he painted the starboard side of the USS Bismarck.

(29) Every boy painted part of the boats. Jerry painted part of the USS Bismarck, Elaine painted the USS Maine and USS Eisenhower, ...

- See also (30), which seems to be OK if we do not know which boat the winch came from

(30) This winch must be part of Columbus' boats (but I'm not sure which one)

- We also observe homogeneity-like effects for these partitives; see again plurals

(31) No New Yorkers came to the party. #Only Jerry did.

- The contradiction in (32) should not appear if a sub-ship part is not in the set denoted by the partitive, yet it does

(32) No part of the boats {was/were} painted. #Jerry (only) painted half of USS Ticonderoga.

- This data shows that plural partitives *can* mean part of an individual in that plurality; this is transparent when certain felicity conditions are met
- SP and FTS accounts, which say that the extension of a partitive DP.PL only includes integrated wholes and their sums, have no way of accounting for the above
- Oddly, though, partitive DP.PL does not pass the implicature cancellation test characteristic of more familiar kinds of implicature (cf. *some* ~ *all*)

(33) Jerry painted part of the boats. #In fact, Jerry painted the starboard side of the USS Anchorage.

- This is not necessarily a threat to the presented account; plurals share this trait

(34) George retrieved éclairs from the bin. #In fact, George retrieved one éclair.

5 Conclusion

- Grice (1975): There is a discrepancy between logical operators and their natural language equivalents

It is a commonplace of philosophical logic that there are, or appear to be, divergences in meaning between, on the one hand, at least some of what I shall call the formal devices— $\neg$, $\wedge$, $\vee$, $\supset$, $(\forall x)$, $(\exists x)$, (ιx) (when these are given a standard two-valued interpretation)—and, on the other, what are taken to be their analogues or counterparts in natural language—such expressions as *not*, *and*, *or*, *if*, *all*, *some* (or *at least one*), *the*.

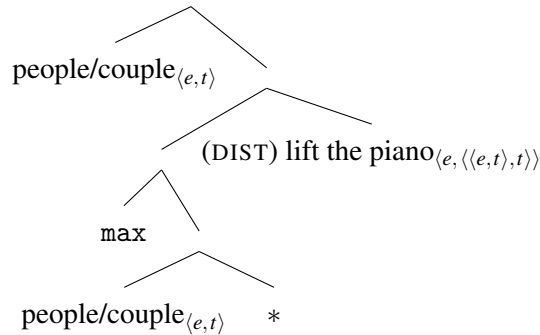
- We can maintain a one-to-one mapping between *some*- $\exists$, *or*- $\vee$, etc. by appealing to an extrasemantic module to give us the strengthened (“conventional”) meaning—we call this module pragmatics
- Leonard and Goodman (1940): via unrestricted sum formation⁴ and the axiom of transitivity, material parts are also part of pluralities
- Given the transitivity problem, this feels wrong; SP and FTS are ways of enriching the meaning of *part* to avoid this outcome
- Channeling Grice, a simpler mereology plus pragmatic inferences get us exactly what we want

⁴Not a universal assumption amongst mereologists (e.g., Bohn 2009), but assumed in CEM.

- Furthermore, §4 bolsters the claim that a simple mereology is the correct treatment for *part*-words; the inclusive interpretation can be accessed given the right felicity conditions
- **Do we need the notion of atom/IW at all?** This is useful for capturing cases of distributivity, counting
- A suggestion from Magri (2008): certain kinds of operations can be quantized with a predicate
- Appealing to predicates instead of atoms/IWs yields an advantage for group nouns—*the couples* and *the people* are counted and distributed to differently, even if they refer to the same plural individual

(35) The people each lifted the piano $\neq$ The couples each lifted the piano

(36) a.



b. $[[\text{DIST lift the piano}]] = \lambda x_e \lambda P_{\langle e,t \rangle} . \forall y \in P . y \sqsubset x \wedge y \text{ lifted the piano}$

- ...but singular group nouns are still compatible with individual-level interpretations, which this approach can't handle (see also pronouns)
- Pending further evidence, I concede that we still need something like atoms/IWs—I submit only that we need not appeal to them to capture facts about *part*-words

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